Electrochimica Acta 305 (2019) 155–163

Contents lists available at ScienceDirect

Electrochimica Acta

journal homepage: www.elsevier.com/locate/electacta

Electron transfer properties of a redox polyelectrolyte based on ferrocenated imidazolium

Valentyn Skrypnik, Dominic Rochefort*

Département de Chimie, Université de Montréal, CP6128 Succ, Centre-Ville, Montréal, Québec, H3C 3J7, Canada

ARTICLE INFO

Article history:
Received 22 November 2018
Received in revised form
21 February 2019
Accepted 28 February 2019
Available online 6 March 2019

Keywords:
Redox poly(ionic liquid)
Electrochemistry
Redox polymer
Composite electrode
Kinetics

ABSTRACT

We report the characterization of a redox polyelectrolyte based on ferrocenated imidazolium and imidazole repeating units to evaluate the impact of the charged structure on the electron transfer rates. *N*-vinyl-*N'*-(methylferrocene)imidazolium chloride monomer was polymerized in the presence of various amounts of vinylimidazole to obtain the electroactive heteropolymer, as the homopolymerization was unsuccessful due to electrostatic repulsion between the imidazolium monomers. NMR was used to determine the ratio of electroactive to neutral monomers in the polymer. The electrochemistry of a thin film of the polymer casted on the surface of glassy carbon electrode was studied in 1 M $NaClO_4$ aqueous electrolyte. Cyclic voltammetry at low scan rates (i.e. $\nu < 0.1\ V\,s^{-1}$) showed a response close to the expected behavior for a surface-confined process without any significant interactions between the ferrocene redox centers, in contrast with poly(vinylferrocene). At high scan rates ($\nu > 1\ V\,s^{-1}$); the behavior was diffusion-controlled. The diffusion coefficient of the ferrocene centers in the imidazolium polymer is $1.7 \times 10^{-9}\ cm^2\,s^{-1}$ which is one order of magnitude higher than poly(vinylferrocene), and the standard rate constant, determined by the Nicholson's approach, was $k^0 = 3.8 \times 10^{-4}\ cm\,s^{-1}$. These high values are explained by favorable interactions of the electrolyte with the charged imidazolium, providing a fast ion diffusion and pairing with ferrocenium. The potential use of this redox polymer in electrochemical systems is demonstrated via the formation of a composite with electrochemically exfoliated graphite to increase polymer loading.

1. Introduction

Redox polymers are polymers with reversible electron exchange capabilities. The oxidation and reduction reactions can take place either on functional groups forming the polymer main chain or via pendant groups tethered to a saturated backbone. Electrically conducting polymers, found in the former class, are highly conjugated systems allowing electronic conduction to take place along the polymer backbone upon oxidation or reduction. In contrast, redox polymers with saturated backbone and pendant electroactive centers are usually not electronic conducting as charges are localized on non-interacting sites. Electronic transport along the polymer chains in these is however possible, although very slow, by exchange between redox centers [1–3]. The properties of the different classes of redox polymers and their application in sensors, catalysis, optoelectronics, and energy storage and conversion have been reviewed in several reports [4–7]. The present publication focuses on polymers with pendant electroactive centers and these will be referred therein as redox polymers.

Of all the electroactive centers used in redox polymers, ferrocene (Fc) is certainly the most studied. The simplest polymeric structure incorporating Fc is poly(vinylferrocene), PVF (Fig. 1). It represents an ideal example of redox polymer because of the well-known electrochemistry of Fc, its high redox activity and stability, allowing to understand the differences in diffusion and electron transfer between its use a solute and as a film. Much of the current knowledge on electron transfer of redox polymer films coated on electrodes comes from the work of Bard and Murray's respective groups on PVF in a series of papers on the matter [8–11]. The incorporation of ferrocene in a PVF film usually results in a decrease in the diffusion coefficient D by a factor of 10^4. The heterogeneous rate constant, k^0, is about 10^2 times lower in the polymer film. It has been noted however that the value for $k^0D^{-1/2}$ for Fc in the polymer is similar to that in solution, suggesting that the solvent properties affecting electron transfer kinetics, being similar in both adsorbed and dissolved cases, are more important than the impact of

* Corresponding author.
E-mail address: dominic.rochefort@umontreal.ca (D. Rochefort).

https://doi.org/10.1016/j.electacta.2019.02.119

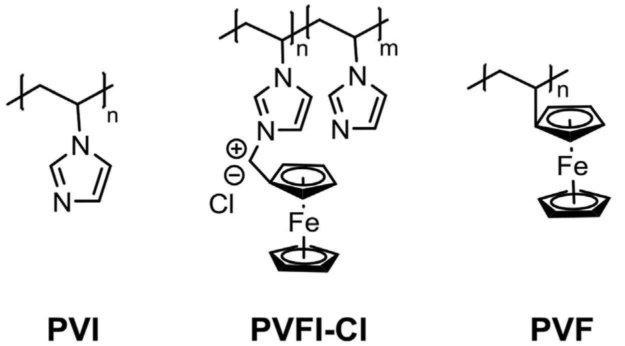

Fig. 1. Chemical structure and acronyms of the polymers used in this study.

covalently linking the redox center. Interestingly, this observation is not limited to covalently bound redox moieties, as films of electrochemically-inert poly(vinylpyridine) loaded with $[Fe(CN)_6]^{4-/3-}$ also show the same trend for k^0 and D, with a similar $k^0D^{-1/2}$ as compared to the electroactive couple in solution [12]. Ferrocene is also of interest due to its rich and well-established chemistry, allowing its incorporation on various polymeric structures. As a few examples, copolymers of metacrylate or ethylene oxide bearing ferrocene in a repeating unit have been used to control the polymer's solubility and morphology while maintaining the redox properties and provide a control on the macroscopic architecture [13,14]. Recently, ferrocene was incorporated in a polymer of ionic liquid via the polymerization of a metacryloyloxy ethyl ferrocenemethyl imidazolium cationic monomer with a bis(trifluoromethylsulfonyl)imide counter ion [15]. Polymer of ionic liquids, or poly(ionic liquids), are macromolecules with repeating units based on a cation/anion pair forming an ionic liquid. These structures have been studied for a decade now, yet it remains unclear how the ionic liquid nature of their monomer results in a behavior different from other polyelectrolytes. The motivation behind their use is to retain the stability properties of ionic liquids without the disadvantage of a liquid phase in energy storage devices for instance. Reviews on the topic can be found presenting the structure, properties and applications of poly(ionic liquids) [16,17]. Redox poly(ionic liquids) however have started to emerge more recently. These are based on the polymerization of an ionic liquid structure containing a redox moiety [18–22]. The possibility of adding electron transfer ability to a polymer with the electrochemical stability of ionic liquids is appealing for energy storage applications and Mecerreyes et al. reported on anthraquinone- and TEMPO-based poly(ionic liquids) for use in lithium batteries, fuel cells, metal-air batteries, and electrolytes in organic redox flow batteries [23]. Interestingly, Randriamahazaka et al. demonstrated that an electrode modified with their poly(ionic liquid) modified with ferrocene could be reversibly oxidized and reduced in a pure solvent in the absence of any supporting electrolyte [15]. They claimed that the cations and anions found in the immediate environment of the ferrocenyl moiety act as self-supporting electrolyte to allow electron transfer. In addition, they demonstrated that the reaction kinetics of their poly(ionic liquid) were much less solvent dependent than for neutral ferrocene-based polymer due to a faster rate of counterion migration in the former. In their case, the electrode was modified by grafting a brush-type polymer on glassy carbon, yielding a layer of ca. 16 nm thick. A significant fraction of ferrocene centers in such films are located at a sufficiently small distance from the electrode (i.e. 5 nm and below) to transfer electrons trough tunnelling. They mentioned that this transfer mechanism might play a significant role in the overall kinetics [15]. The electrochemical behavior of significantly thicker layers of redox poly(ionic liquid) structures should be therefore evaluated to understand the kinetics of ferrocene centers located at longer ranges.

To do so, we developed a new polyelectrolyte poly(*N*-vinyl-*N′*-(methylferrocene)imidazolium) chloride (PVFI-Cl, Fig. 1) based on ferrocenated imidazolium repetition units that can be easily synthesized from available starting materials (Fig. 2). The polymer was casted on glassy carbon electrodes from an aqueous solution to study its electrochemical response by cyclic voltammetry over a wide range of scanning rates. The casting technique yielded significantly thicker films than the grafting approach. The diffusion coefficient and electron transfer rate constant of the ferrocenyl moieties in the film were determined and compared with PVF. Finally, the polymer was incorporated as a composite with electrochemically exfoliated graphite to increase the loading and demonstrate a potential application in energy storage devices.

2. Experimental

2.1. Chemicals

N,*N*-dimethylaminomethylferrocene (96%), *N*-vinylimidazole (≥99%), and 2,2′-azobis(2-methylpropionitrile) (98%, AIBN) were purchased from Sigma-Aldrich. AIBN was recrystallized in methanol and other regents were used as received. All solvents were reagent grade, except the HPLC MeOH purchased from Fisher Scientific. All salts used for electrolyte preparation were high purity grade and they were purchased from Sigma-Aldrich. Poly(-vinylferrocene) was purchased from Polyscience Inc. Deionized water with resistivity of 15 MΩ cm was used in washing and solution preparation. EC-G was prepared according to the approach published earlier [24], but using graphite foil (0.5 mm thick, 99.8%) as electrode instead of conductive carbon tape coated with graphite flakes. Argon (99,998%) gas was obtained from Praxair. The glassy carbon electrode (1.5 mm radius) and Ag/AgCl reference electrode were obtained from BASi, and a platinum wire 99,997% (metal basis) from Alfa Aesar was used for counter electrode. Pyrolytic Graphite sheet purity 99.90% (thickness 0.017 mm ± 0.005 mm, d = 2.1 g/cm^3) was obtained from MTI Corporation.

2.2. Synthesis

2.2.1. Synthesis of o-acetylmethylferrocenyl

The following protocol was adapted from a previously published procedure [25]. *N*,*N*-dimethylaminomethylferrocene (**I**, 35.2 g, 145 mmol) was dissolved in neat acetic anhydride (300 mL, 3.17 mol). The reaction flask was heated in an oil bath of reflux for 3 h. The mixture was then cooled in an ice bath and it was hydrolyzed with the slow addition of saturated aqueous Na_2CO_3 (300 mL). The organic phase was extracted with diethyl ether (3 × 300 mL). The organic phases were combined and concentrated by evaporating 2/3 of the diethyl ether under reduced pressure. The remaining organic concentrate was washed with saturated $NaHCO_3$ until no evolution of CO_2 was observed. It was then dried over anhydrous $MgSO_4$ and filtered. After evaporating the remaining diethyl ether under reduced pressure, an orange solid was isolated.

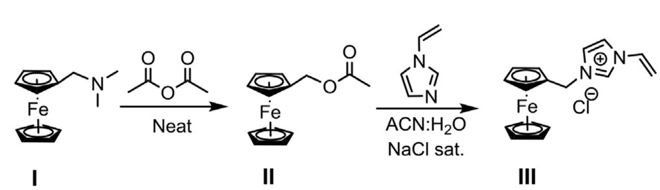

Fig. 2. Synthetic pathway for the VyImCH$_2$FcCl monomer (**III**).

o-acetylmethylferrocenyl (**II**) was dried under vacuum at 65 °C for 12 h to yield a crystalline orange solid (34,1 g, 91%).

^{1}H NMR (400 MHz, $CDCl_3$): δ (ppm) 2.04 (s, 3H, C**H_3**$COOCH_2Fc$), 4.16 (s, 5H, C_5**H_5** not bonded), 4.18 (t, 2H, $^{3}J_{H-H} = 1.8$ Hz), 4.27 (t, 2H, $^{3}J_{H-H} = 1.8$ Hz C_5**H_4** bonded both with 4.18), 4.89 (s, 2H, CH_3COOC**H_2**Fc).

HR MS (ESI) *m*/*z*: [M^+] (for $C_{13}H_{14}FeO_2$, found: 258.0341 g/mol calculated: 258.0343 g/mol additional fragment found: 199.0214 g/mol).

2.2.2. Synthesis of N-vinyl-N′-methylferrocenylimidazolium chloride

o-Acetylmethylferrocene (**II**, 10.0 g, 116 mmol) was dissolved in acetonitrile (100 mL). To the orange solution, was added *N*-vinylimidazole (13.0 mL, 138 mmol) under stirring. A saturated aqueous solution of NaCl (100 mL, 35.7 g) was then added to the resulting solution. The turbid mixture was heated in oil bath under reflux for 12 h. The red solution was cooled and filtered over a glass frit to remove the insoluble salts. The solution was washed with Et_2O to remove the unreacted *o*-acetylmethylferrocenyl and the aqueous phase was removed under reduced pressure. The red oil was mixed with other salts (NaCl and NaOAc) were recovered. The product was solubilised in $CHCl_3$ and filtered twice over a glass frit to remove the insoluble salt. The organic phase was concentrated and it was purified by flash chromatography using aluminum oxide phase. Acetone was used as the first eluent to elute the unreacted *N*-vinylimidazole and MeOH was used to elute *N*-vinyl-*N*′-methylferrocenylimidazolium chloride. The isolated product was dried under reduced pressure and the product is dried in vacuum oven at 65 °C for 12 h to afford the title compound as red solid (16.4 g, 65%). ^{1}H NMR (400 MHz, DMSO-d_6): δ (ppm) 4.23 (s, 5H, C_5**H_5** not bonded), 4.25 (t, 2H, $^{3}J_{H-H} = 1.8$ Hz), 4.48 (t, 2H, $^{3}J_{H-H} = 1.8$ Hz, C_5**H_4** bonded both with 4.25), 5.21 (s, 2H, $H_2C{=}CH$-ImC**H_2**FcCl), 5.39 (dd, 1H, $^{3}J_{H-H} = 8.8$ Hz (cis), $^{2}J_{H-H} = 2.3$ Hz (gem)), 5.98 (dd, 1H, $^{3}J_{H-H} = 15.6$ Hz (trans), $^{2}J_{H-H} = 2.3$ Hz (gem), **H_2**C=CH-ImCH_2FcCl coupled both with 5.39), 7.31 (dd, 1H, $^{3}J_{H-H} = 8.8$ Hz (cis), $^{3}J_{H-H} = 15.6$ Hz (trans), $H_2C{=}C$**H**-ImCH_2FcCl), 7.92 (m, 1H, N^+–C**H**=CH–N), 8.20 (m, 1H, N^+–CH=C**H**–N), 9.63 (m, 1H, N^+–C**H**–N). HR MS (ESI) *m*/*z*: [M^+] (for $C_{17}H_{12}FeN_2$, found: 293.0735 g/mol calculated: 293.0741 g/mol, also found 199.0207 g/mol and dimer form $C_{32}H_{34}ClFe_2N_4$, found 621.1170 g/mol calculated 621.1170 g/mol).

2.2.3. Synthesis of poly(N-vinyl-N′-(methylferrocenyl)imidazolium-co-N-vinylimidazole chloride)

N-vinyl-*N*′-methylferrocenylimidazolium chloride (**III**, 1.50 g, 4.56 mmol) was dissolved in $CHCl_3$ (10 mL). To the red solution, was added *N*-vinylimidazole (41 μL, 0.45 mmol) followed by the addition of AIBN (300.3 mg, 2 wt%). The reaction media was then purged with argon 15 min. The reaction flask was transferred to an oil bath and heated under reflux at 65 °C. After 24 h, the $CHCl_3$ is evaporated under reduced pressure. The solids were solubilised in MeOH (10 mL) with the slow addition of MilliQ (15 MΩ cm) water (40 mL). The light yellow/orange solution was then transferred to a dialysis membrane (regenerated cellulose, ~25 cm long and 4.3 cm large) and it was dialysed against MilliQ water (20 L) for 2 days. The water was changed once. The tubes were recovered and the water was freeze dried the polymer as a fluffy orange powder (1.01 g). ^{1}H NMR (700 MHz, DMSO-d_6): δ (ppm): 7.60-6.20 (br, aromatic imidazolium protons), 5.23-4.66 (br, 3 broad sigulets, $H_2C{=}CH$-ImC**H_2**FcCl), 4.56-4.33 (br, C_5**H_4**), 4.33-4.08 (br, C_5**H_5** not linked), 2.44-1.70 (br, appears as 2 broad peaks, R**H_2**C-CHR-ImC**H_2**FcCl).

2.2.4. Film preparation

Films were prepared by dropcasting solution of PVFI-Cl, EC-G:PVFI-Cl, PVF, or EC-G:PVF on glassy carbon electrode. The casted solution was prepared by dissolving 5.0 mg of the polymer (PVFI-Cl or PVF) in 5.00 mL of MeOH or THF. For the suspensions of composite, the weight was adjusted to also obtain 1 mg mL^{-1} of polymer in the solvent. A 1 μL volume of the solution was deposited on the electrode surface and the solvent evaporated with a gentle steam of argon gaz. Prior to each cast, the glassy carbon electrode was polished with a MicroPolish 1 and 0.05 μm alumina suspension and sonicated for 5 min in MilliQ water. The film volume was determined by measuring its thickness with a DektakXT profilometer of the PVFI-Cl film of on a flat glassy carbon substrate, after equilibrating the film in the 1 M $NaClO_4$ electrolyte for 30 min. The thickness and volume values were obtained by averaging the data for three different depositions of the same volume of solution.

2.3. Characterization

2.3.1. FTIR and Raman spectroscopy

Spectra were recorded on a Bruker ALPHA-P FTIR spectrometer with an ATR unit with 64 scans, 4 cm^{-1} resolution in the range of 4000–400 cm^{-1}. Raman spectra were acquired on a Renishaw RM 3000 with a 514.5 nm green laser with 30 s of irradiation time with 10% of the laser power. The polymer powder sample was pressed into a pellet for these measurements.

2.3.2. Cyclic voltammetry

Cyclic voltammetry was performed on a VMP3 BioLogic Science Instruments using a typical three-electrode setup (working electrode: film-coated vitreous carbon electrode; counter electrode: Pt wire; reference: Ag/AgCl).

2.3.3. CP-MAS

Standard CP-MAS solid-state NMR experiments were performed on a Bruker AVANCE 600WB with a conventional CP pulse sequence [26] on a powder of the polymer compacted in the sample holder. Individual samples were introduced and compacted in 2.5 mm zirconia rotor and the measurements were done at 12 kHz. The contact time was set to 0.5 ms and the recycling delay was 2 s. The number of scans were 6 k for PVFI-Cl, 150 k for EC-G and 100 k for EC-G:PVFI-Cl.

2.3.4. SEM-EDX

All micrographs were obtained on the JOEL JSM7600F electronic microscope equipped with a field effect canon. Electrons were accelerated with a 5 kV acceleration potential. Electron dispersive X-ray was employed for Fe mapping of the composite material.

2.3.5. XPS

XPS data was obtained from a VG ESCALAB 3 MKII using Mg Kα radiation (hν = 1253.6 eV, 300 W). The size of the analyzed surface was 2 mm × 3 mm. The electrons were collected at a perpendicular take off angle resulting in a maximal analyzed depth of 10 nm. For survey scans, an energy step size of 1 eV was used with a pass energy of 100 eV. For high resolution scans, the energy steps were set to 0.05 eV with a pass energy of 20 eV. The pressure in the analysis chamber was kept below 5×10^{-9} torr during data acquisition. The data were collected and processed with an Avantage 4.2 software.

3. Results and discussion

3.1. Polymer synthesis and characterization

Acetylmethylferrocene (**II**) was firstly obtained from the reaction of commercially available dimethylaminomethyleneferrocene

(**I**) in neat acetic anhydride [25]. Subsequently, **II** was subjected to a nucleophilic substitution with vinylimidazole in a mixture of acetonitrile and a saturated aqueous solution of sodium chloride to obtain the *N*-vinyl-*N*′-(methylferrocene)imidazolium chloride monomer (VyImCH$_2$FcCl: **III**). Mass spectrometry was used to confirm the success of the synthesis, as well as proton NMR and IR analysis which will be discussed below. The VyImCH$_2$FcX monomer was detected both in aggregates (at higher masses) and also its isolated form, as commonly seen in ESI-MS spectra of ionic liquids [27]. Mass spectrometry data for all derived methylferrocene **I**, **II** and **III** compounds also showed the presence of a 199.0207 g/mol mass. This peak is indicative of the presence of α-ferrocenylmethylcarbocation likely as a result of the ionization step. It has been already observed in the mass spectra of other α-substituted alkylferrocenes [28]. The 293.07357 *m*/*z* signal observed corresponds to the VyImCH$_2$FcCl monomer which was polymerized in the next step.

Initial attempts to homopolymerize **III** via free radical polymerization with azobisisobutyronitrile (AIBN) was unsuccessful, despite having been reported by Mecerreyes et al. for the polymerization of alkyl-substituted vinylimidazolium [29]. The proposed mechanism is shown in Fig. S1. We believe that the combination of the bulkier ferrocene and the electrostatic repulsion hindered the polymerization. Amajjahe and Ritter demonstrated that an increasing distance between the *N*-vinyl-*N*′-alkylimidazolium and its associated anion leads to strong electrostatic repulsion between the cations [30]. The presence of ferrocene close to the imidazolium likely weakens its association with the chloride anion thereby favoring electrostatic repulsion between the imidazolium units. An alternative to the AIBN approach to polymerize ionic liquids with fluorinated counterions using cobalt-mediated controlled radical polymerization was reported by Detrembleur et al. [31] To avoid a possible contamination with residual electroactive cobalt complexes in the polymer, this strategy was not employed here. Therefore, vinylimidazole was added to the VyImCH$_2$FcCl monomer in various amounts to increase spacing between the imidazolium units and promote polymerization (Fig. 3). Using this approach, several heteropolymers were obtained (Table 1). The polymerization was successful when 10 mol% of vinylimidazole was added to the reaction mixture.

The ^{1}H NMR spectra of the VyImCH$_2$FcCl monomer and of the heteropolymer PVFI-Cl (obtained with equimolar amounts of VyImCH$_2$FcCl and vinylimidazole) are shown in Fig. 4. The NMR spectra of the PVFI-Cl using other ratios are presented in Fig. S2 of the Supporting Information. All peaks were assigned to their respective protons using available NMR data (detailed assignment in the experimental section). As expected, most of the peaks are shifted to higher field because of the positive charge on the imidazolium, confirming that the shielding constant is sensitive to the interaction between cations and anions [32]. Fig. 4 also shows the ^{1}H NMR spectrum of poly(*N*-vinylimidazole) (PVI) for comparative purposes. PVI was prepared according to the protocol described by Zhao et al. [33], although the reaction was performed in chloroform instead of dimethylformamide. First, significant differences between spectra **A** and **B** is the broadening of all peaks. The additional broadening of the peaks to spectrum **B** compared in spectrum **C** is most probably related to the charge found on the ionic polymer. As previously discussed by Allen et al. [34], molecular diffusion and

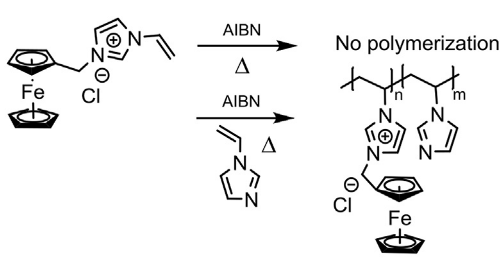

Fig. 3. Polymerization scheme of VyImCH$_2$FcCl to PVFI-Cl (IV).

Table 1
Comparison of the equivalent amounts of VyImCH$_2$FcCl and vinylimidazole monomers added in the preparation of PVFI-Cl and those found in the isolated polymer (by integration of NMR peaks).

VyImCH$_2$FcCl to vinylimidazole ratio (added)	n: m ratio found in polymer IV
100% VyImCH$_2$FcCl	No polymerization
50 : 1	No polymerization
10 : 1	6.8 : 1
5 : 1	3.8 : 1
1 : 1	1.5 : 1
0.1 : 1	0.5 : 1

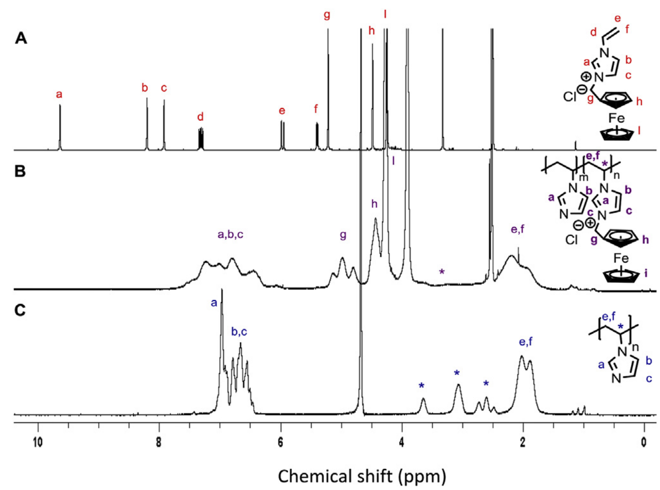

Fig. 4. NMR spectra of VyImFcCl (A), PVFI-Cl (B) and PVI (C). H$_\alpha$ and H$_\beta$ are poorly resolved on this figure and these appear as peak h at 4.48 ppm and 4.25 ppm, respectively. Peaks at 3.25 ppm, 3.9 ppm and 4.25 ppm are due to DMSO-d_6, H_2O occlusion, and D_2O, respectively. Asterisks indicate protons responsible for tactility peak splitting.

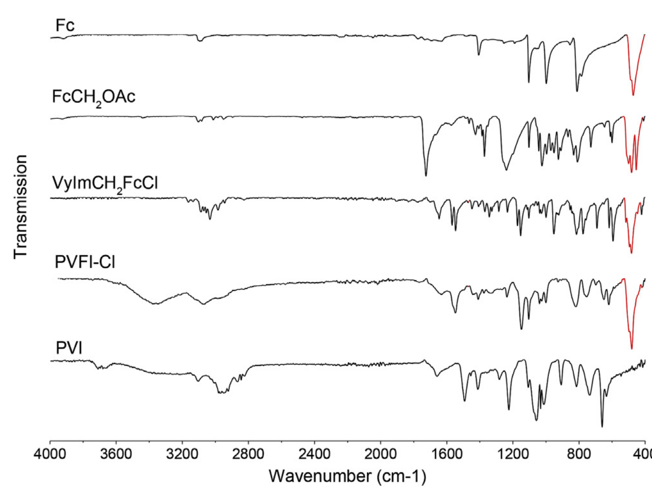

Fig. 5. FTIR spectra of ferrocene, subsequent compound functionalized with Fc and PVI. The band associated with the ferrocene ring tilt mode is marked in red.

rotational motion contributes to relaxation process so do charge density and acidic nature of a bond. Those conclusions were established on T_1 relaxation experiments. Indeed, ionic polymers are highly charged and they have restricted degree of motion, broad peaks are therefore to be expected from typical ^{1}H NMR experiments. Another important feature distinguishing the ionic monomer and the polymer is the signals for the aromatic protons a,*b* and c. These are displaced to lower fields and appeared between 6 ppm and 7.5 ppm, in the same chemical shift range as PVI aromatic protons. Oddly, the protons associated with the tacticity in PVI identified by (*) are merged with the baseline for PVFI-Cl. The presence of the actual signal was established from nuclear overhauser effect (NOE) experiments by irradiating the methylene (e and f) on the polymer backbone (Supporting Information Fig. S3).

In the case of PVI, Dambatta et al. [35] demonstrated that the peaks at 2.6 ppm, 3.2 ppm and 3.6 ppm are expected for the uncontrolled radical polymerization of *N*-vinylimidazole. These signals are associated to the methine protons and relate to the tacticity of the polymer chain. The relative intensities of the component follows Bernoullian statistics with the probability of the meso placement *Pm* ~0.46 (21:50:29). The steric triads were to isotactic (3.6 ppm), heterotactic (3.2 ppm) and syndiotactic (2.8 ppm) on the basis of ^{1}H and ^{13}C NMR analyses. In the case of PVFI-Cl, the peak is exceedingly broad over (more than 1.5 ppm) and very weak, to the extent of merging with the baseline. Lastly, it is important to discuss the nature of the three peaks at 5.13 ppm, 4.97 ppm and 4.79 ppm (**g**). Those signals appear as three singulets for the methylene, as shown by the HSQC-DEPT results (Supporting Information Fig. S4). The three slightly different magnetic environments found are most likely arising from the cation-π interaction [36].

The analysis of the ^{1}H NMR spectra in Fig. 4 and S1 allowed to evaluate the relative fractions of each unit type in the heteropolymer. The results of Table 1 were obtained by comparing the integration of the triplet peak g and the sum of peaks a, b and c between 6 and 8 ppm. As stated above, the homopolymerization was unsuccessful and we found that the heteropolymerization occurred when at least 10% of monomers were vinylimidazole. The sample with an n: m ratio of 1.5 : 1 was chosen for the rest of the experiments (referred as PVFI-Cl).

The chemical structure of ferrocenyl in each intermediate compound **I-III** as well in PVFI-Cl was verified by IR spectroscopy. Fig. 5 and Table 2, show the IR-active vibrations modes ferrocene, acetate methylferrocene, VyImCH$_2$FcCl, PVFI-Cl, and PVI. The assignations presented in Table 2 were done based on published work based on DFT calculations on ferrocene and substituted analogous compounds [37,38]. The propagation of the ring tilt mode occurring at ~ 480 cm^{-1} for a A_2'' symmetry is spread to Fc derived compound and is retained in the rigid polymeric form. Ring tilt vibration is observed at 496 cm^{-1}. No significant changes are noticed for the Fe-Cp stretching when comparing all compounds from ferrocene to PVFI-Cl. The band for the out-of-plane CH motion has only a slight decrease of 2 cm^{-1} for FcCH$_2$OAc and a small increase of 4 cm^{-1} and 7 cm^{-1} respectively for VyImCH$_2$FcCl and PVFI-Cl compared to the crystalline ferrocene. Such changes suggest a slight increase in the Fe-Cp distance as the excitation energy decreases. Ferrocene has 2 active IR modes for D_{5d} symmetry. D_{5d} has 57 vibrational degrees of freedom with 15 Raman and 10 in infrared fundamental vibrational frequencies. The appearance of multiple bond frequencies around 1410 cm^{-1} and 1560 cm^{-1} is one evidence for equivalent multiple bonds, which result from the delocalization of the π-electrons [39]. Multiple peaks of weak intensity appear in the region around 1700 cm^{-1}. These are assigned to overtones and combination bands based on the previous interpretations of benzene ring and its substituted derivatives [40]. The overtones also the FcCH$_2$OAc and they are more resolved in VyImCH$_2$FcCl. Addition vibrational modes were observed for the imidazole ring modes upon protonation to imidazolium [41–43]. A shift in the position in ring modes (C—C, C—N and C—H) was noted but no significant change on the stretching of the backbone C—H. The appearance of a new peak of medium intensity at 1172 cm^{-1} is associated with the protonated imidazolium ring. In our case, a new sharp peak appeared at 1148 cm^{-1} for PVFI-Cl. The appearance of new peaks in the region 1110 cm^{-1}-1200 cm^{-1} on the polymer spectra are consistent with the presence of the imidazolium ring, based on a comparison of *N*-vinylimidazole and VyImCH$_2$FcCl (1153 cm^{-1} and 1172 cm^{-1}). The broad peak that spans from 3300 cm^{-1} to 3400 cm^{-1} is assigned to a small amount of water occlusion in the polymer. Altogether, these observations point to fact that the reactions used for the monomer preparation and the polymerization did not affect the ferrocenyl moiety and that it should remain electroactive.

3.2. Electrochemistry of PVFI-Cl redox polymer

In order to ensure electrochemical measurements to be as reliable as possible, the volume of the polymer film deposited by solution casting on the glassy carbon substrate was determined from 4 independent depositions, using a profilometer as described in the experimental section. The confidence interval (95%) for the film volume is determined to be $(4.9 \pm 0.2) \times 10^{-6}$ g, or 4%. Cyclic voltammetry experiments were carried out in order to assess the electroactive behavior of PVFI-Cl and evaluate its differences with PVF. A 1 mg mL^{-1} solution of the PVFI-Cl (1:1 ratio) was prepared in a mixture of water and methanol. A similar solution was prepared for PVF but using THF as the solvent. A 1 μL volume of the solution was pipetted on the surface of a glassy carbon disk and the solvent evaporated. This procedure yields highly reproducible films as evidenced by the small differences in CVs obtained for three

Table 2
Assignment of the vibrations to the absorption bands of the FTIR spectra in Fig. 5.

	Wavenumber (cm^{-1})			
	Fc	FcCH$_2$OAc	VyImCH$_2$FcCl	PVFI-Cl
Fe-Cp stretching	472	482	483	480
out of plane CH	812	810	816	819
bending CCH	999	996; 1024	952; 1000	819; 1000
Breathing	1103	1103	1103	1103
C—N stretching	–	–	1153	1149
C=C, C5H5 stretching	1407	1372; 1426	1407	1409
C—O stretching	–	1239	–	–
C=C, Vinyl stretching	–	–	1547; 1646	1548; 1632
C=O stretching	–	1727	–	–
CH stretching	3094	2950; 3012; 3102	2943; 2983; 3032; 3086	2950; 3073; 3370

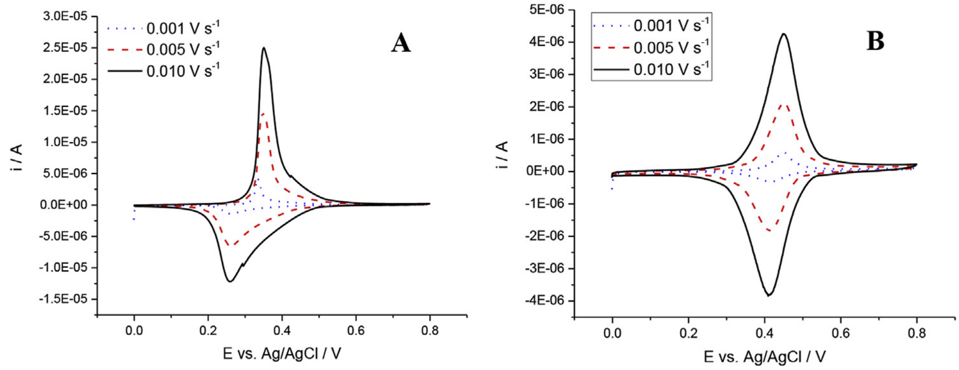

Fig. 6. CV of PVF (A) and PVFI-Cl (B) film casted on glassy carbon electrode in 1 M $NaClO_4$ aqueous electrolyte measured at low scan rates.

different electrodes shown in Fig. S5. Fig. 6 shows a comparison of CVs recorded at low (i.e. <0.010 V s^{-1}) scan rates of equivalent weights of each polymer in the form of a drop-cast film on the surface of vitreous carbon electrodes.

In both cases, the electrolyte was an aqueous solution of 1 M $NaClO_4$. This was chosen because in other electrolytes, namely sodium salts of sulfate, nitrate and acetate, PVFI-Cl dissolved rapidly upon oxidation. This was evidenced by a rapid decrease in the peak current upon the first redox cycles in these electrolytes while the current remained constant in 1 M $NaClO_4$ (Fig. S6 of the Supporting Information). Oxidation-induced dissolution is favored for electrolyte anions that provide hydrophilic ion pairs with ferrocenium. This was reported for a film of poly(vinylferrocene) in a 1 M H_2SO_4 aqueous electrolyte [44]. Inzelt and Szabo studied the effect of the nature and the concentration of counter ions on the electrochemistry of PVF and showed that oxidized ferrocenium repeating units are stabilized with the association of the perchlorate anions to provide more hydrophobic ion pairs [45]. For instance, ClO_4^- anions provided a stable cyclic voltammetric response while Cl^- cause a dissolution of the film from the electrode surface. While the initial PVFI-Cl counterion is Cl^- for synthesis purpose, it is believed that the anion is exchanged with ClO_4^- during the experiment to provide a stable CV response. The comparison of the electrochemical response of PVF and PVFI-Cl, two major differences are noted when comparing the CV of Fig. 6. First, the peak potential values of PVFI-Cl are shifted towards positive values vs. those for PVF due to the inductive effect of the *N*-methylimidazolium. An increase in potential of +200 mV has been reported for ferrocene derivative of imidazolium, 1-(ferrocenylmethyl)-3-methylimidazolium, redox ionic liquids compared to unmodified ferrocene [46]. The shapes of the curves are also significantly different. The distorted peaks and dissimilarity in shape between the anodic and cathodic parts of the curve for PVF at low scan rates is explained by interactions within the film as electrons are transferred between neighboring ferrocenes [47]. There is no evidence for such interactions in the PVFI-Cl heteropolymer as the CV displays the typical shape for non-interacting center. The addition of vinylimidazole increases the distance between redox centers, making them unable to efficiently exchange charge. Log($i_{p,a}$)-log(ν) curve was drawn for PVFI-Cl using potential scan rates of 0.1 V s^{-1} and below, and provided a slope of 0.91 (see Fig. S7 of the Supporting Information). This value indicates a predominance of surface-confined process (slope = 1) for the PVFI-Cl, in concordance with the CV curve shape. A more precise analysis of the peak current dependency on scan rate is provided in Fig. S8 and S9 for low and high scan rates, respectively. A linear function with scan rate is obtained at low values (from 0.001 to 0.02 V s^{-1}) as expected for a confined process. At high rates, the linear function is obtained with the square root of scan rate demonstrating a diffusional behavior between 0.5 and 20 V s^{-1}. This range will therefore be used to determine the rate constant. The oxidation of ferrocene in PVF is also expected to be a surface-confined process but the peak current is strongly influenced by the interaction between neighboring sites.

Under the assumption, justified above, that there is no significant electron transfer by hopping between the redox centers of the PVFI-Cl we can use classical approaches to evaluate the electron transfer rate constant. Using such approach, the redox centers are considered as independent diffusing entities, although their diffusion rate should be significantly lower than in solution because they are restricted to the polymer structure. The determination of the standard rate constants k^0 using the Nicholson's approach for a polymer film deposited on the surface of an electrode requires knowing the diffusion coefficient of the redox centers, which in turn requires their concentration in the film [48]. The number of electroactive ferrocenyl moieties was determined by integrating the area of the anodic or cathodic current from cyclic voltammograms recorded at very low scan rate (e.g. 0.001 V s^{-1} in Fig. 6 B). The thickness of the film used for the electrochemical measurements was 0.24 μm, yielding a concentration of (3.6 ± 0.2) x 10^{-3} mol cm^{-3} of electrochemically active ferrocene. The diffusion coefficient was then measured by chronoamperometry from the slope of the current plotted as a function of $t^{-1/2}$ (Fig. S10, Supporting Information) using Cottrell's equation. The value obtained of (1.7 ± 0.1) x 10^{-9} cm^2 s^{-1} is significantly higher than the diffusion of ferrocene centers reported for a PVF film in a similar electrolyte (D = (2.6 ± 0.2) x 10^{-10} cm^2 s^{-1} in 0.5 M $LiClO_4$) [45]. In order to determine k^0, cyclic voltammograms were recorded between 0.5 and 20 V s^{-1}. The curves displayed in Fig. 7A show a diffusion-controlled process in clear contrast with the curves in Fig. 6 recorded at low scan rates. The oxidation and reduction peak potential gradually move towards positive and negative values with scan rate and the ΔE_p was used to calculated k^0 (Nicholson's approach). The cell resistance was determined to be 29 Ω using Electrochemical Impedance Spectroscopy. This uncompensated resistance was used to calculate the iR drop at each scan rate using the peak current and the ΔE_p was manually corrected accordingly. The heterogeneous electron transfer rate found, (3.8 ± 0.5) x 10^{-4} cm s^{-1}, is ten times higher than that of a PVF film in a $TBABF_4$/MeCN electrolyte in which the ferrocene centers have a diffusion coefficient similar to our PVFI-Cl [47]. The confidence interval provided for the values given above were calculated from a set of 3 independent electrodes with a 95% confidence level. The role of the ionic liquid structure on the high rate constant and diffusion coefficient values can be explained on the basis of the observations of Randriamahazaka et al. made for a thin layer of ferrocene-imidazolium brush polymer [15]. As the oxidation process of

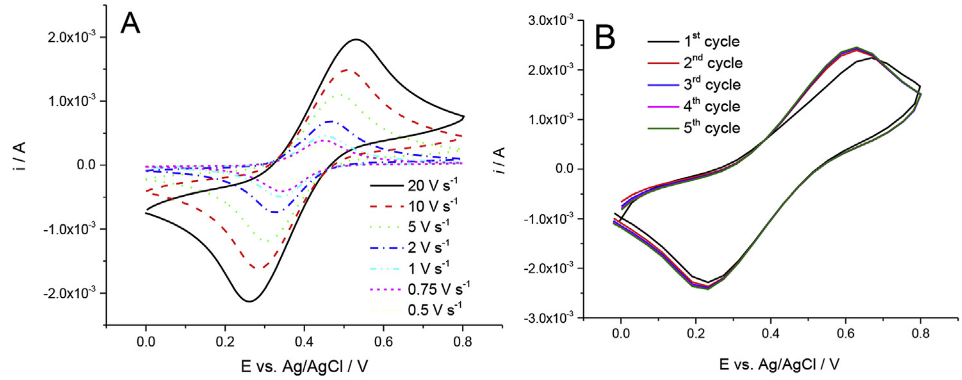

Fig. 7. Effect of the scan rate on the CV of PVFI-Cl film on glassy carbon electrode in 1M $NaClO_4$ electrolyte (A) and first 5 cycles recorded at 100 V s^{-1} (B). The peak potential difference values in A was measured at each scan rate and used to calculate electron transfer rate constant. The 3rd cycle is shown at each scan rate. The non-zero current at the beginning of the first scan in (B) is explained by the fact that this curve was recorded right after those shown in (A) without holding the potential at E_i.

ferrocene immobilized on an electrode surface is strongly dependent on ion pairing with the counterions from the electrolyte [49], a fast ion pairing is promoted by favorable interaction between the polymer and the electrolyte. Films of polymers based on ferrocene-imidazolium repeating units present both a good wettability (90° contact angle in aqueous PBS) [15] and charges in the proximity of the redox centers. These features result in a high electrolyte take-up in the film and fast counterion transport to the redox site, explaining the fast electron transfer kinetics observed here and by Randriamahazaka et al. Fig. 7B shows that a diffusion-controlled process is maintained even up to 100 V s^{-1}.

3.3. PVFI-Cl-carbon composite electrode

Composite films of polyvinylferrocene redox-active polymers and carbon allotropes have been proposed to improve properties of several electrochemical systems for energy storage and sensing [50–53]. As unconjugated polymers lack the ability to rapidly transfer electrons with the underlying electrode surface, the addition of carbon nanotubes or graphitic particles allow long range electronic conduction in thick films which contain higher amounts of active materials. In order to assess the potential use of PVFI-Cl in such composites, we incorporated our polymer in electrochemically exfoliated graphite (EC-G) particles prepared according to a published methodology. [24] Such EC-G was easily dispersed in MeOH along with PVFI-Cl and the powder obtained after solvent evaporation was used for further characterization. The composite contains equivalent weights of carbon and polymer and is denoted as EC-G:PVFI-Cl (1:1). The presence and integrity of the polymer was firstly confirmed using cross-polarisation magic angle spinning (CP-MAS) NMR (Fig. 8).

Under the same measurement conditions, it is possible to see the different carbons in the PVFI-Cl both in the pure polymer and in the composite (denoted EC-G:PVFI-Cl). The peaks observed in the same region for composite suggest that the carbons from ferrocene, imidazolium and polymeric aliphatic backbone are all present in

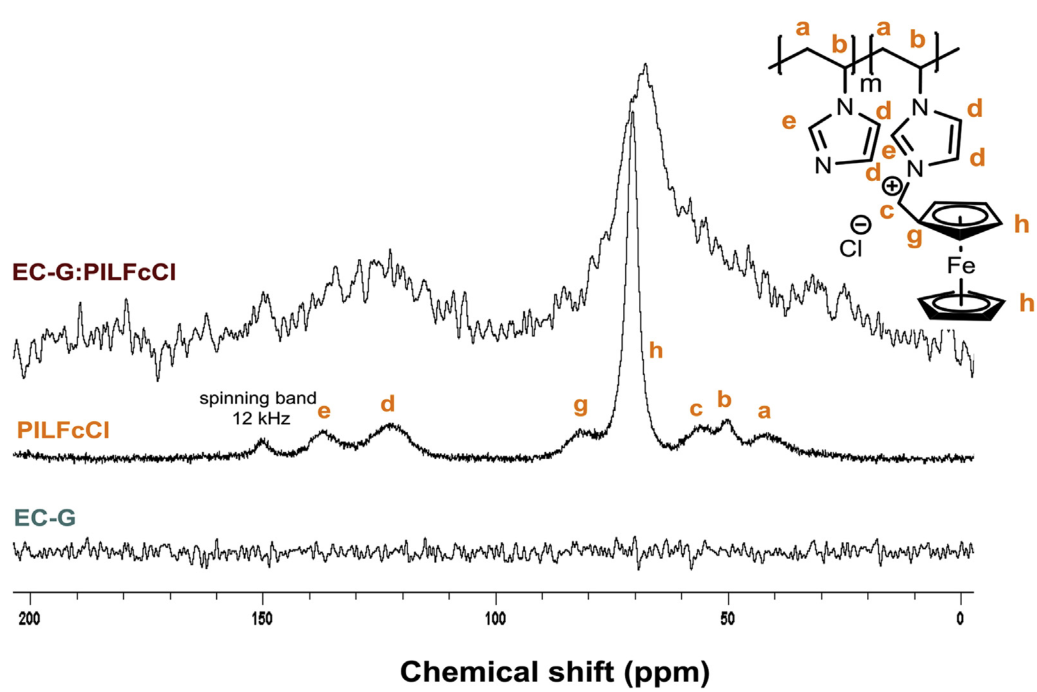

Fig. 8. CP-MAS ^{13}C NMR spectrums for PVFI-Cl, EC-G and the composite. (EC-G:PVFI-Cl) recorded at a 12 kHz spinning frequency.

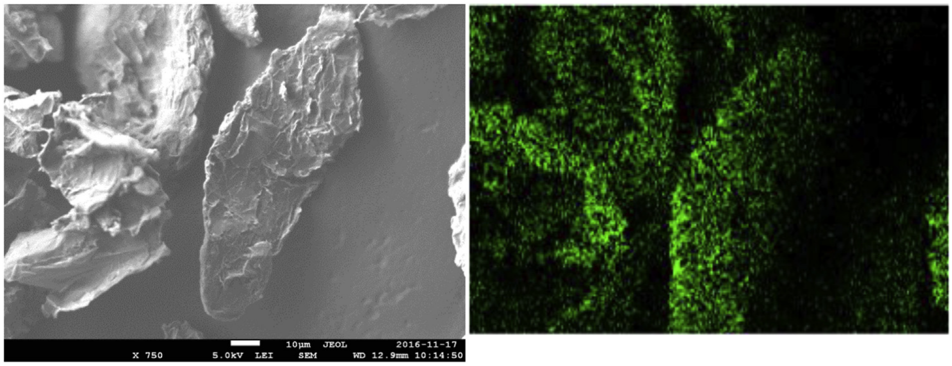

Fig. 9. SEM micrograph and corresponding EDX (Fe L) image of the EC-G:PVFI-Cl (1:1) composite. The scale bar shown represents 10 μm.

the composite material. The signal intensity decreases significantly between the pure polymer and the composite, resulting in a poorer peak resolution. Yet, the results indicate that cross polarisation occurs from proton to linked carbons and allows to confirm the presence of aliphatic backbone carbons (b,a) as well as the aromatic imidazolium (e,d) and ferrocene carbons (g,h). XPS and EDX were used to confirm the presence of iron on the surface of the graphite particles. Fe(II) from ferrocene is seen in the high resolution spectra of the Fe *2p* region, shown in Fig. S11 (Supporting Information). Scanning electron microscopy (SEM) image and the corresponding EDX mapping (Fe L) are shown in Fig. 9. EDX shows than the polymer covers most of the graphite particles in the form of small aggregates. To evaluate its electrochemical behavior, the composite was dispersed in MeOH and casted on the surface of a glassy carbon electrode. Fig. 10 shows the CV curve (3rd cycle, 10 V s^{-1}) for the EC-G:PVFI-Cl composite. For comparison, a CV recorded under the same conditions but for the PVFI-Cl film without any carbon is also shown. The amount of composite was adjusted so the quantity of polymer is identical in both experiments. The composite electrode yielded a significantly higher current (factor of 3.7) than the pure polymer electrode despite having the same number of redox-active sites, showing that the addition of carbon allows an efficient electronic conducting with more active sites. Smaller carbon structures such as carbon nanotubes could likely increase further the activity of the polymer by providing a better dispersion. Finally, Fig. S13 shows the current evolution over 100 cycles recorded at 0.05 V s^{-1}. The 18% decrease between the first and last cycle is likely due to a slow dissolution of the polymer from the composite. Ways to improve stability such as cross-linking or covalent binding of the polymer to the carbon structure will be explored in future works.

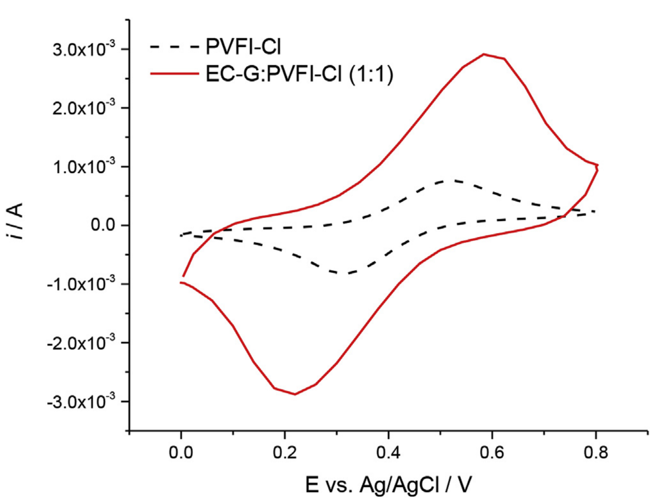

Fig. 10. A: Comparison of the CV (1 M $NaClO_4$ at 10 V s^{-1}) for a film of pure PVFI-Cl and a composite of the polymer with electrochemically-exfoliated graphite EC-G:PVFI-Cl (1:1). The quantity of composite was adjusted to have identical amounts of PVFI-Cl on each electrode.

4. Conclusions

Increased electron transfer rates have been shown for very thin films of ferrocene-imidazolium poly(ionic liquids). The increase in kinetics was explained by the favorable interactions between the ionic liquid structure and the electrolyte, as well as with the highly charged nature of the polymer. In very thin films (i.e. below 20 nm) such as those obtained by grafting the polymer on an electrode, tunnelling might play a significant role in the overall observed kinetics. In order to demonstrate that the imidazolium ionic liquid structure is responsible for the enhance electron transfer, we prepared a redox-active polymer based on an ionic liquid structure via the polymerization of a ferrocenated vinylimidazolium salt. While the homopolymerization was impossible to achieve because of electrostatic repulsion, a heteropolymer with *N*-vinylimidazole was obtained. The redox-active polymer (with chloride as counter-ion; PVFI-Cl) is soluble in mixtures of water and methanol, allowing its casting on the surface of a glassy carbon electrode in thicker films (over 200 nm). Cyclic voltammetry shows that PVFI-Cl has a very different behavior than polyvinylferrocene (PVF) at low scanning rates. While PVF shows strong interactions between the ferrocene centers, those in PVFI-Cl behaved independently, likely because of the spacing between the redox centers. Below 0.1 V s^{-1}, the current response fitted to a surface-confined process rather than being diffusion limited. The behavior becomes diffusion limited for scan rates above 0.5 V s^{-1} and the film showed a high electrochemical reversibility up to 100 V s^{-1}. The heterogeneous rate constant of for PVFI-Cl, (3.8 ± 0.5) x 10^{-4} cm s^{-1}, is higher than PVF and likely explained by a fast ion pairing between ferrocenium and the electrolyte anions due to the densely charged polymer. This observation confirms the hypothesis that was proposed earlier for very thin films of redox-active poly(ionic liquids), showing the potential of these structures for energy storage devices.

Acknowledgments

This work was supported by the Discovery Grant program of the Natural Sciences and Engineering Research Council of Canada. The authors thank Bruno Gélinas, Ph.D., for his input in designing the synthesis protocols and Cedric Malveau, Ph.D., from the Université de Montréal Regional Center of NMR Spectroscopy for his assistance with the solid-state NMR measurements.

Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.electacta.2019.02.119.

References

[1] N.A. Surridge, J.C. Jernigan, E.F. Dalton, R.P. Buck, M. Watanabe, H. Zhang, M. Pinkerton, T.T. Wooster, M.L. Longmire, Electron self-exchange dynamics between redox sites in polymers, Faraday Discuss. Chem. Soc. 88 (1989) 1–17.
[2] R.H. Terrill, R.W. Murray, Electron hopping transport in electrochemically active, molecular mixed valent materials, Mol. Electron. (1997) 215–239.
[3] A. Akhoury, L. Bromberg, T.A. Hatton, Interplay of electron hopping and bounded diffusion during charge transport in redox polymer electrodes, J. Phys. Chem. B 117 (2013) 333–342.
[4] M.M. Barsan, M.E. Ghica, C.M.A. Brett, Electrochemical sensors and biosensors based on redox polymer/carbon nanotube modified electrodes: a review, Anal. Chim. Acta 881 (2015) 1–23.
[5] N. Casado, G. Hernández, H. Sardon, D. Mecerreyes, Current trends in redox polymers for energy and medicine, Prog. Polym. Sci. 52 (2016) 107–135.
[6] R. Gracia, D. Mecerreyes, Polymers with redox properties: materials for batteries, biosensors and more, Polym. Chem. 4 (2013) 2206–2214.
[7] C.P. Andrieux, J.M. Saveant, Catalysis at redox polymer coated electrodes, Tech. Chem. (N. Y.) 22 (1992) 207–270.
[8] J.B. Flanagan, S. Margel, A.J. Bard, F.C. Anson, Electron transfer to and from molecules containing multiple, noninteracting redox centers. Electrochemical oxidation of poly(vinylferrocene), J. Am. Chem. Soc. 100 (1978) 4248–4253.
[9] P.J. Peerce, A.J. Bard, Polymer films on electrodes: Part II. Film structure and mechanism of electron transfer with electrodeposited poly(vinylferrocene), J. Electroanal. Chem. Interfacial Electrochem. 112 (1980) 97–115.
[10] J. Leddy, A.J. Bard, Polymer films on electrodes: Part XII. Chronoamperometric and rotating disk electrode determination of the mechanism of mass transport through poly(vinyl ferrocene) films, J. Electroanal. Chem. Interfacial Electrochem. 153 (1983) 223–242.
[11] P. Daum, R.W. Murray, Charge-transfer diffusion rates and activity relationships during oxidation and reduction of plasma-polymerized vinylferrocene films, J. Phys. Chem. 85 (1981) 389–396.
[12] N. Oyama, T. Ohsaka, M. Kaneko, K. Sato, H. Matsuda, Electrode-kinetics of the $Fe(CN)_6^{4-/3-}$ and $Fe(CN)_5^{3-/2-}$ complexes confined to polymer film on graphite surfaces, J. Am. Chem. Soc. 105 (1983) 6003–6008.
[13] C. Tonhauser, A. Alkan, M. Schömer, C. Dingels, S. Ritz, V. Mailänder, H. Frey, F.R. Wurm, Ferrocenyl glycidyl ether: a versatile ferrocene monomer for copolymerization with ethylene oxide to water-soluble, thermoresponsive copolymers, Macromolecules 46 (2013) 647–655.
[14] R.H. Staff, M. Gallei, M. Mazurowski, M. Rehahn, R. Berger, K. Landfester, D. Crespy, Patchy nanocapsules of poly(vinylferrocene)-based block copolymers for redox-responsive release, ACS Nano 6 (2012) 9042–9049.
[15] V. Bui-Thi-Tuyet, G. Trippé-Allard, J. Ghilane, H. Randriamahazaka, Surface and electrochemical properties of polymer brush-based redox poly(ionic liquid), ACS Appl. Mater. Interfaces 8 (2016) 28316–28324.
[16] D. Mecerreyes, Polymeric ionic liquids: broadening the properties and applications of polyelectrolytes, Prog. Polym. Sci. 36 (2011) 1629–1648.
[17] J.Y. Yuan, D. Mecerreyes, M. Antonietti, Poly(ionic liquid)s: an update, Prog. Polym. Sci. 38 (2013) 1009–1036.
[18] H.J. Xie, B. Gélinas, D. Rochefort, Electrochemical and physicochemical properties of redox ionic liquids using electroactive anions: influence of alkylimidazolium chain length, Electrochim. Acta 200 (2016) 283–289.
[19] B. Gélinas, D. Rochefort, Synthesis and characterization of an electroactive ionic liquid based on the ferrocenylsulfonyl(trifluoromethylsulfonyl)imide anion, Electrochim. Acta 162 (2015) 36–44.
[20] B. Gelinas, J.C. Forgie, D. Rochefort, Conductivity and electrochemistry of ferrocenyl-imidazolium redox ionic liquids with different alkyl chain lengths, J. Electrochem. Soc. 161 (2014) H161–H165.
[21] J.C. Forgie, D. Rochefort, Electroactive imidazolium salts based on 1,4-dimethoxybenzene redox groups: synthesis and electrochemical characterisation, RSC Adv. 3 (2013) 12035–12038.
[22] E. Mourad, L. Coustan, P. Lannelongue, D. Zigah, A. Mehdi, A. Vioux, S.A. Freunberger, F. Favier, O. Fontaine, Biredox ionic liquids with solid-like redox density in the liquid state for high-energy supercapacitors, Nat. Mater. 16 (2017) 446–453.
[23] G. Hernandez, M. Isik, D. Mantione, A. Pendashteh, P. Navalpotro, D. Shanmukaraj, R. Marcilla, D. Mecerreyes, Redox-active poly(ionic liquid)s as active materials for energy storage applications, J. Mater. Chem. A 5 (2017) 16231–16240.
[24] K. Parvez, Z.-S. Wu, R. Li, X. Liu, R. Graf, X. Feng, K. Müllen, Exfoliation of graphite into graphene in aqueous solutions of inorganic salts, J. Am. Chem. Soc. 136 (2014) 6083–6091.
[25] M. Lamač, I. Císařová, P. Štěpnička, Synthesis and structural characterization of rac-2-[(diphenylphosphino)methyl]ferrocenecarboxylic acid, its selected derivatives and some rhodium complexes, J. Organomet. Chem. 690 (2005) 4285–4301.
[26] J. Schaefer, E.O. Stejskal, Carbon-13 nuclear magnetic resonance of polymers spinning at the magic angle, J. Am. Chem. Soc. 98 (1976) 1031–1032.
[27] P.J. Dyson, I. Khalaila, S. Luettgen, J.S. McIndoe, D. Zhao, Direct probe electrospray (and nanospray) ionization mass spectrometry of neat ionic liquids, Chem. Commun. (2004) 2204–2205.
[28] W.E. Watts, Cheminform abstract: ferrocenylcarbocations and related species, Chem. Informationsdienst 10 (1979).
[29] R. Marcilla, J. Alberto Blazquez, J. Rodriguez, J.A. Pomposo, D. Mecerreyes, Tuning the solubility of polymerized ionic liquids by simple anion-exchange reactions, J. Polym. Sci. Part A Polym. Chem. 42 (2004) 208–212.
[30] S. Amajjahe, H. Ritter, Anion complexation of vinylimidazolium salts and its influence on polymerization, Macromolecules 41 (2008) 716–718.
[31] D. Cordella, A. Kermagoret, A. Debuigne, C. Jérôme, D. Mecerreyes, M. Isik, D. Taton, C. Detrembleur, All poly(ionic liquid)-based block copolymers by sequential controlled radical copolymerization of vinylimidazolium monomers, Macromolecules 48 (2015) 5230–5243.
[32] R. Caminiti, L. Gontrani, The Structure of Ionic Liquids, Springer, 2013.
[33] W. Zhao, Y. Tang, J. Xi, J. Kong, Functionalized graphene sheets with poly(ionic liquid)s and high adsorption capacity of anionic dyes, Appl. Surf. Sci. 326 (2015) 276–284.
[34] C.J. Allen, J. Hwang, R. Kautz, S. Mukerjee, E.J. Plichta, M.A. Hendrickson, K.M. Abraham, Oxygen reduction reactions in ionic liquids and the formulation of a general orr mechanism for li–air batteries, J. Phys. Chem. C 116 (2012) 20755–20764.
[35] B.B. Dambatta, J.R. Ebdon, T.N. Huckerby, Unusual influences of temperature and medium on the tacticity of radically polymerised poly(n-vinyl imidazole), Eur. Polym. J. 20 (1984) 645–652.
[36] J.C. Ma, D.A. Dougherty, The cation–π interaction, Chem. Rev. 97 (1997) 1303–1324.
[37] T.P. Gryaznova, S.A. Katsyuba, V.A. Milyukov, O.G. Sinyashin, DFT study of substitution effect on the geometry, IR spectra, spin state and energetic stability of the ferrocenes and their pentaphospholyl analogues, J. Organomet. Chem. 695 (2010) 2586–2595.
[38] E.R. Lippincott, R.D. Nelson, The vibrational spectra and structure of ferrocene and ruthenocene, Spectrochim. Acta 10 (1958) 307–329.
[39] M. Cais, J.J. Dannenberg, A. Einsenstadt, M.I. Levenberg, J.H. Richards, Nuclear magnetic resonance spectra of ferrocenyl carbonium ions, Tetrahedron Lett. 7 (1966) 1695–1701.
[40] C.W. Young, R.B. DuVall, N. Wright, Characterization of benzene ring substitution by infrared spectra, Anal. Chem. 23 (1951) 709–714.
[41] A.-M. Bellocq, C. Perchard, A. Novak, M.-L. Josien, Spectres de vibration de l'imidazole, de l'imidazole (d)-1, de l'imidazole (d3)-2,4,5 et de l'imidazole (d4), J. Chim. Phys. 62 (1965) 1334–1343.
[42] C. Perchard, A.-M. Bellocq, A. Novak, Spectres de vibration de l'imidazole, de l'imidazole (d)-1, de l'imidazole (d3)-2,4,5 et de l'imidazole (d4), J. Chim. Phys. 62 (1965) 1344–1358.
[43] J.L. Lippert, J.A. Robertson, J.R. Havens, J.S. Tan, Structural studies of poly(n-vinylimidazole) complexes by infrared and Raman spectroscopy, Macromolecules 18 (1985) 63–67.
[44] T.B. Issa, P. Singh, M.V. Baker, Electrochemistry of poly(vinylferrocene) modified electrodes in aqueous acidic media, J. Power Sources 140 (2005) 388–391.
[45] G. Inzelt, L. Szabo, The effect of the nature and the concentration of counter ions on the electrochemistry of poly(vinylferrocene) polymer film electrodes, Electrochim. Acta 31 (1986) 1381–1387.
[46] O. Fontaine, C. Lagrost, J. Ghilane, P. Martin, G. Trippé, C. Fave, J.C. Lacroix, P. Hapiot, H.N. Randriamahazaka, Mass transport and heterogeneous electron transfer of a ferrocene derivative in a room-temperature ionic liquid, J. Electroanal. Chem. 632 (2009) 88–96.
[47] J. Leddy, A.J. Bard, Polymer films on electrodes: Part XVIII. Determination of heterogeneous electron transfer kinetics at poly(vinylferrocene) and nafion/$Ru(bpy)_2^{+3}$ polymer-modified electrodes by convolution voltammetry, J. Electroanal. Chem. Interfacial Electrochem. 189 (1985) 203–219.
[48] T. Ohsaka, S. Kunimura, N. Oyama, Electrode kinetics of poly (o-aminophenol) film prepared by electro-oxidative polymerization of *o*-aminophenol and its electrochromic properties, Electrochim. Acta 33 (1988) 639–645.
[49] K.-L. Nguyen, E.R. Dionne, A. Badia, Redox-controlled ion-pairing association of anionic surfactant to ferrocene-terminated self-assembled monolayers, Langmuir 31 (2015) 6385–6394.
[50] X.W. Mao, E.H. Yan, G.C. Rutledge, T.A. Hatton, Enhanced redox transformation efficiency in unconjugated electroactive polymer/carbon nanotube hybrids, Chem. Mater. 28 (2016) 543–548.
[51] M.T. Sulak, E. Erhan, B. Keskinler, Amperometric phenol biosensor based on horseradish peroxidase entrapped PVF and PPY composite film coated GC electrode, Appl. Biochem. Biotechnol. 160 (2010) 856–867.
[52] W.D. Tian, X.W. Mao, P. Brown, G.C. Rutledge, T.A. Hatton, Electrochemically nanostructured polyvinylferrocene/polypyrrole hybrids with synergy for energy storage, Adv. Funct. Mater. 25 (2015) 4803–4813.
[53] X. Mao, F. Simeon, D.S. Achilleos, G.C. Rutledge, T.A. Hatton, Metallocene/carbon hybrids prepared by a solution process for supercapacitor applications, J. Mater. Chem. A 1 (2013) 13120–13127.